

Sorting. Workers not only decide whether to adopt chatbots, but also which firms to sort into. Since all adopters receive the same net benefit from encouragement, $\hat{\alpha}_1$, workers effectively face a choice between sorting into encouraged firms and adopting AI chatbots, or sorting into non-encouraged firms and not adopting. The condition for sorting into encouraged firms is given by:

$$E_{j(i)} = \mathbf{1} [\log Y_{i1}(1) - \log Y_{i0}(0) \geq C_{i1}] \quad (24)$$

$$= \mathbf{1} [\hat{\alpha}_0 + \hat{\alpha}_1 + \hat{\beta} \times i + \log W_1 - \log W_0 \geq 0]. \quad (25)$$

This implies a strict cutoff rule for worker sorting:

$$E_{j(i)} = \mathbf{1} [i \leq i^*], \quad \text{with} \quad i^* = -\frac{1}{\hat{\beta}}(\hat{\alpha}_0 + \hat{\alpha}_1 + \log W_1 - \log W_0). \quad (26)$$

A.2.3 Equilibrium

An equilibrium is a set of encouragement policies $\{E_j\}_{j \in \mathcal{J}}$, skill prices $\{W_j\}_{j \in \mathcal{J}}$, allocation of workers to firms $\{\mathcal{I}_j\}_{j \in \mathcal{J}}$, and adoption decisions $\{\{A_{ij}\}_{i \in \mathcal{I}_j}\}_{j \in \mathcal{J}}$, such that firms encourage AI chatbots and hire workers to maximize profits (Equations (15) and (16)), workers sort into firms and adopt AI chatbots to maximize utility (Equations (23) and (26)), and labor and product markets clear (Equations (13) and (14)).

The initial equilibrium without AI chatbots is simple: because all firms are symmetric, they each employ an equal share of the baseline human capital stock. The following paragraphs characterize the equilibrium when AI chatbots are available.

Sorting. Worker sorting across firms when AI chatbots are available is straightforward: Assuming adopters receive higher benefits from encouraged firms ($\hat{\alpha}_1 \geq 0$), the equilibrium features full segregation, with all adopters in encouraged firms, all non-adopters in non-encouraged firms, or both. We focus on the empirically relevant case in which some, but not all, firms encourage AI chatbot use, and some, but not all, workers adopt.

Skill prices. Equilibrium skill prices adjust to equalize the supply and demand of human capital across encouraged and non-encouraged employers, which in turn depend on workers' sorting and adoption decision rules as well as firms' encouragement and hiring policies.

Let κ denote the encouragement cost of the marginal adopting employer. From Equation (19), equilibrium skill price differentials satisfy:

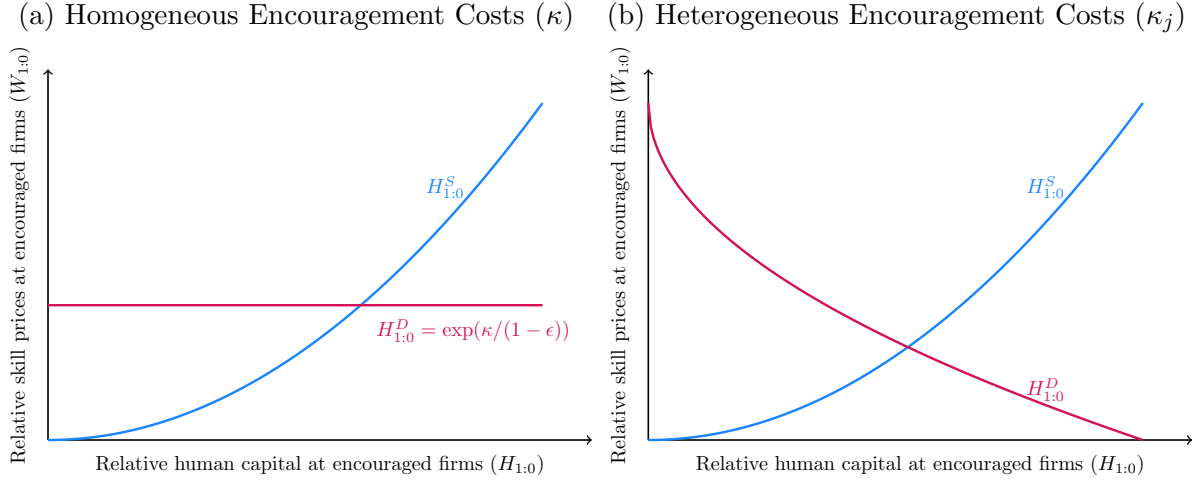
$$\log W_1 - \log W_0 = \frac{\kappa}{1 - \epsilon}. \quad (27)$$

Thus, the cost of employer encouragement is priced into skill differentials: workers are not rewarded for these costly investments.

A simple case arises when all employers face the same encouragement cost, $\kappa_j = \kappa$ for all j . In this case, all firms are marginal to encouraging use, and the share of encouraged employers adjusts to match the relative supply of human capital by adopting workers. For analytical simplicity, the model predictions in Section A.2.4 focus first on the case with homogeneous encouragement costs. Section A.3.2 extends the analysis to the general case with heterogeneous costs.

Figure A.2 illustrates the inverse aggregate supply and demand curves for relative human capital at encouraged firms. Panel (a) depicts the case where firms face the same encouragement cost κ , while Panel (b) illustrates the more general case with heterogeneous encouragement costs.

Figure A.2: Equilibrium Skill Price Differentials



Notes: This figure illustrates the labor market equilibrium across encouraged and non-encouraged firms. Equilibrium skill price differentials, $W_{1:0} = W_1/W_0$, are determined by the aggregate supply $H_{1:0}^S = H_1^S/H_0^S$ and aggregate demand $H_{1:0}^D = H_1^D/H_0^D$ for relative human capital at encouraged vs. non-encouraged firms.

A.2.4 Model Predictions

In this section, we use the model to derive predictions for how the arrival of AI chatbots affects labor market outcomes, including worker earnings, occupational mobility, and sorting across firms.

Worker earnings. Combining Equations (9)–(10) yields the following expression for the log change in earnings of worker i at firm j :

$$d \log Y_{ij} = \underbrace{d \log W_j}_{\text{skill-price effect}} + \underbrace{A_i B_{ij}}_{\text{productivity effect}} \quad (28)$$

$$= d \log W_0 + E_j \frac{\kappa}{1-\epsilon} + A_i(\alpha_{b0} + \beta_b \times i) + E_j A_i \alpha_{b1}, \quad (29)$$

Equation (29) yields three key predictions about how AI chatbots affect worker earnings.

1. Workers who do not use chatbots experience an earnings loss at encouraged firms:

$$\log Y_{i1}(A_i = 0) - \log Y_{i0}(A_i = 0) = -\frac{\kappa}{\epsilon - 1} \leq 0 \quad (30)$$

2. Holding encouragement status fixed, adoption increases worker earnings by the

productivity effect:

$$\log Y_{iE}(A_i = 1) - \log Y_{iE}(A_i = 0) = \alpha_{b0} + \alpha_{b1} \times E + \beta_b \times i \quad (31)$$

3. The impact of employer encouragement on adopter earnings is theoretically ambiguous, as it depends on the relative magnitudes of the productivity and skill price effects:

$$\log Y_{i1}(A_i = 1) - \log Y_{i0}(A_i = 1) = \alpha_{b1} - \frac{\kappa}{\epsilon - 1} \geq 0. \quad (32)$$

Even if employer encouragement enhances the productivity benefits of chatbots ($\alpha_{b1} > 0$, as documented in Section D.2.3), these gains need not translate into higher earnings for adopters. When encouragement is costly (high κ), product demand is less elastic (low ϵ), and productivity gains are modest (low α_{b1}), employer encouragement may reduce adopter earnings. Adopters may still prefer to sort into encouraged firms if encouragement sufficiently reduces adoption costs (i.e., if $\alpha_{c1} < \alpha_{b1} - \frac{\kappa}{\epsilon - 1}$). However, these benefits will not be reflected in their earnings.

Occupational mobility. If chatbots create greater benefits B_{ij} for newcomers to an occupation, we will observe a dynamic correlation between chatbot adoption and occupational switching (as documented in Section 3.2).²²

Worker-firm sorting. Following Equation (26), AI chatbots should trigger a reallocation of workers who would like to adopt chatbots (low- i types) into firms that encourage use ($E = 1$).

Worker outcomes. Figure A.3 summarizes how AI chatbots affect workers' sorting, adoption, and earnings. The gray line highlights the equilibrium sorting and adoption

²²This model prediction reflects an effect running from occupational choice to chatbot adoption. The causality may also run the other way, however, with chatbots prompting workers to move into new occupations. The theoretical framework abstracts from this occupational-choice margin to keep the analysis tractable.